605.486 – Android Development

Module 20
Property Animation

JavaBean Properties

- Defined by public get/set methods
- Data doesn't matter! Only method names!
  - `public Type getName()`
  - `public void setName(Type name)`
  - Property name -> drop the "get"/"set" and lowercase the name
    - `setFirstName()`
      - defines a writable property called "firstName"

Property Example

```
public class Person {
  private String firstName;

  public String getFirstName() {
    return firstName;
  }

  public void setFirstName(String firstName) {
    this.firstName = firstName;
  }
}
```

Bound Property Example

```
public class Person {
  private String firstName;
  private PropertyChangeSupport pcs = new PropertyChangeSupport(this);

  public void addPropertyChangeListener(PropertyChangeListener listener) {
    pcs.addPropertyChangeListener(listener);
  }
  public void removePropertyChangeListener(PropertyChangeListener listener) {
    pcs.removePropertyChangeListener(listener);
  }

  public String getFirstName() {
    return firstName;
  }

  public void setFirstName(String firstName) {
    String oldValue = this.firstName;
    this.firstName = firstName;
    pcs.firePropertyChange("firstName", oldValue, firstName);
  }
}
```

Constrained Properties

- Fire VetoableChangeEvents
  - Modified Observer Pattern
  - Observers throw exceptions to “veto” change
- Don’t use!

Property Animation

- Don’t think about graphical animation!
- ObjectAnimator changes property values
  - `ObjectAnimator.ofInt(bean, property, start, end)`
  - Duration
  - Interpolator
  - Repeat count
  - Repeat Mode

Example 1 - Animate Age

- Layout – displays age
- Person class – simple name and age properties (logs changes)
- Activity – starts animator

Example 2 - Bound Property

- Adds listeners
- Activity listens to changes and updates UI
- Stopping animation -> animator.cancel()
- Reverse animation -> animator.reverse()

Example 3 - Moving Text

- ViewPropertyAnimator
  - Simple but limited
- Rotate and translate text

Example 4 - Animation Set

- Combine animations into a set
  - Start/Cancel at once
- Allows reverse

Example 5 - Shifting Panels

- Switching panels by animating left/right
- Note that animations won't actually start until UI thread has control again!
- AnimatorListener
  - On animation end
    - Hide layout
    - Stop listening!

Example 6 - Adding Alpha Animation

- Fade layout that is being hidden

Example 7 - Flip Layouts

- Use rotation Y
- Rotate first view from 0 to 90
  - On animation end
    - Rotate second view from -90 to 0
- (Idea from ListFlipper demo – implemented differently)

Example 8 - Sliding Layout From Side

- Useful for sidebar (navigation, etc)

- Main Layout
  - Two fragments
  - Activity hides and shows one

Example 9 - Using Gesture to Slide

- Add GestureDetector
  - Only care about “fling”
  - Note: tricky to perform on emulator!

- Note: must delegate onTouchEvent!